ANALYTICAL COMPOSITE MECHANICS

Predicting effective Young's modulus

Six model families | Reproducible Python project

Objective. Calculate effective elastic stiffness from constituent properties, volume fractions and explicitly stated geometry assumptions.

Implemented scope. Linear elastic, perfectly bonded, isotropic solid phases with spherical inclusions. The program includes simple mixture formulas, Eshelby/dilute, Mori-Tanaka, classical self-consistent, Hashin-Shtrikman bounds and a separate Halpin-Tsai parameter demonstration.

Data status. The shipped inputs are hypothetical teaching cases. This report demonstrates analytical behavior and implementation verification; it does not claim experimental accuracy or a measured biocomposite result.

Delivered component	Included
Executable workflow	Single cases, material-file inputs, fraction sweeps
Results	CSV tables, PNG/PDF graphs, run manifest
Verification	22 tests; 120 seeded material cases
Study material	Theory, assumptions, classification, references and viva guide

Woven architectures, real-fiber anisotropy and experimental validation remain extensions. Their background is discussed without claiming that the spherical model predicts those structures.

01 / Background and learning path

What is a composite?

A composite combines constituents with distinct roles. The matrix surrounds the reinforcement and transfers load through the interface. Young's modulus E describes elastic stiffness: it is not a prediction of failure strength.

Classification	Examples
By matrix	Polymer, metal, ceramic
By reinforcement	Particles, short fibers, continuous fibers
By architecture	Unidirectional, woven, random reinforcement, laminate
Woven examples	Plain, twill, satin, basket, warp rib, weft rib

Natural fibers such as flax, jute and hemp motivate biocomposite research. A real natural-fiber case needs sourced properties and a defensible geometry and anisotropy model. The current data are not measurements of these fibers. [R7-R9]

Learning sequence

1. Stress, strain and the meaning of elastic modulus.

2. Phase volume fractions, interfaces and loading direction.

3. Simple mixtures; then Poisson ratio, bulk and shear modulus.

4. Concentration tensors, Eshelby and mean-field assumptions.

5. Bounds, numerical residuals, plots and limits of validation.

Why woven geometry matters

Warp/weft fractions, yarn directions, crimp and matrix pockets influence woven stiffness. These cannot be recovered from Ef, Em, Vf and Vm alone.

02 / Inputs, baselines and Eshelby

For a void-free two-phase system, Vf + Vm = 1. Use Vf=0.30 for 30%, not 30. Phase Poisson ratios are required by the three-dimensional models.

```
K = E/[3(1-2 nu)] G = E/[2(1+nu)]
E = 9KG/(3K+G) nu = (3K-2G)/[2(3K+G)]
```

Simple mixtures and proper isotropic bounds

```
E_equal_strain = Vf Ef + Vm Em
E_equal_stress = 1/(Vf/Ef + Vm/Em)
```

These scalar idealizations are displayed separately from the 3-D isotropic comparison. The latter averages K and G arithmetically for Voigt and harmonically for Reuss, then converts the pairs to E. [R1]

Spherical Eshelby and dilute stiffness

```
A_dil = [I + S Cm^-1 (Cf-Cm)]^-1
C_dil = Cm + Vf (Cf-Cm) A_dil
```

The actual spherical Eshelby tensor is implemented in orthonormal Kelvin notation, with square-root-of-two scaling of shear components. It is also available as a Cartesian fourth-order tensor. [R2]

Dilute theory ignores inclusion interactions. It appears only for Vf up to 0.1 in the comparison workflow. This chosen cutoff is a teaching restriction, not proof of accuracy. The Vf=1 dilute endpoint is rejected.

03 / Mean-field estimates and bounds

Mori-Tanaka: the matrix as host

```
zeta_m = Gm(9Km+8Gm)/[6(Km+2Gm)]
K_MT = Km + Vf(Kf-Km)/[1+Vm(Kf-Km)/(Km+4Gm/3)]
G_MT = Gm + Vf(Gf-Gm)/[1+Vm(Gf-Gm)/(Gm+zeta_m)]
```

The tensor implementation normalizes the dilute concentration factors to satisfy the average-strain condition. Independent scalar reductions check that implementation. A connected matrix-inclusion topology is assumed. [R3]

Classical self-consistent: the composite as host

```
sum_i Vi(Ki-K)/(Ki+4G/3) = 0
sum_i Vi(Gi-G)/(Gi+zeta(K,G)) = 0
```

K is eliminated analytically at each trial G. Bisection in log(G) solves the remaining equation. Both dimensionless residuals must pass the tolerance check before a result is returned. This is not a finite element calculation or the generalized coated-sphere scheme. [R4]

Hashin-Shtrikman: a permissible range

Lower and upper K/G bounds are computed for well-ordered isotropic phases and converted to E. Phase ordering must hold in both K and G. Cross-ordered cases are explicitly marked unavailable in the tables. [R5]

For ordered isotropic spherical cases, Mori-Tanaka matches the HS boundary associated with its matrix phase. Coincident curves are expected, not a reason to change a model to make the plot look different.

04 / Halpin-Tsai and reproducible execution

A separate semi-empirical demonstration

```
eta = (Ef/Em-1)/(Ef/Em+xi)
E_HT = Em(1+xi eta Vf)/(1-eta Vf)
```

xi depends on geometry, loading and calibration. The shipped xi=2 is a parameter demonstration only. The associated plots are kept separate from the isotropic 3-D estimates. xi=0 recovers the simple Reuss E expression and supplies a useful limiting check. [R6]

Reproduce the entire analysis

```
python -m pip install -r requirements.txt
python main.py --all --report
```

The command creates per-case sweeps, key-fraction CSV tables, PNG/PDF graphs, a run manifest, actual test output and this report. The shipped sweep spans Vf=0 to 0.6; pure-phase endpoints are verified separately.

Use actual inputs

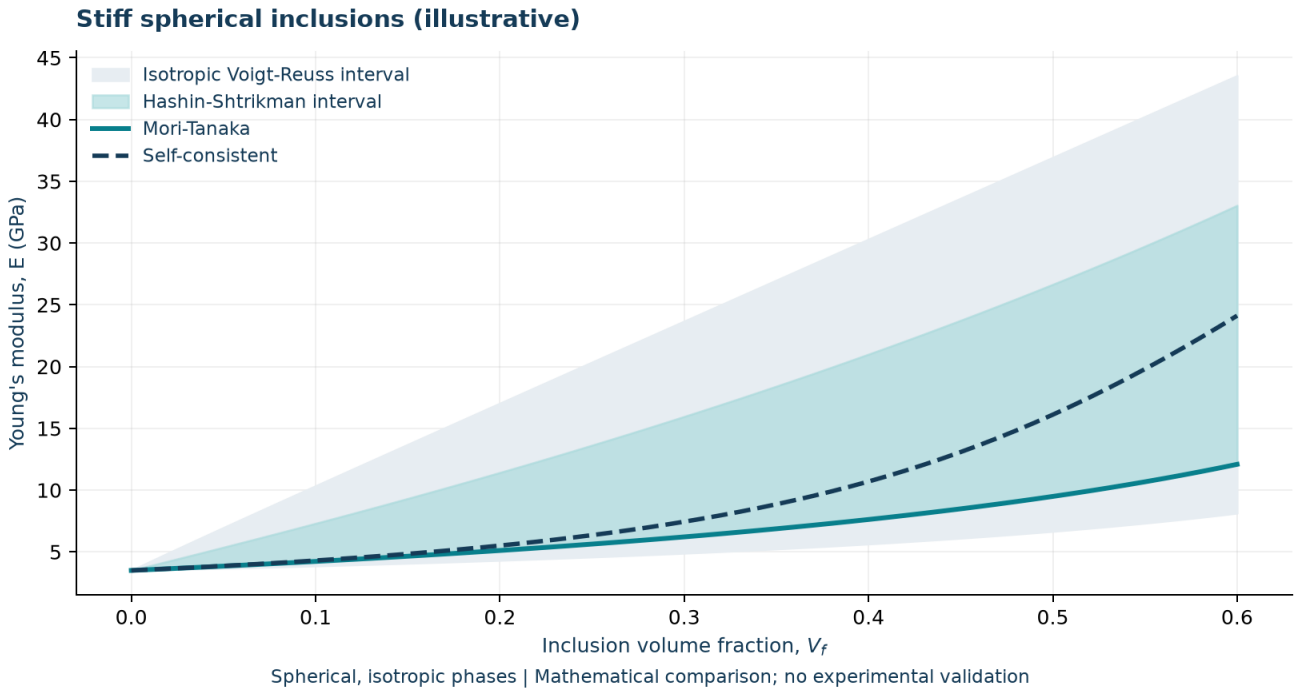
Edit or copy data/materials.json. Supply consistent units, Ef, Em, nu_f, nu_m, a case id, xi and source notes. Then run the selected case or comparison. Do not relabel an illustrative value as measured data.

```
python main.py --case stiff_inclusions --vf 0.05
python main.py --compare --case stiff_inclusions
python main.py --data data/materials.json --list-cases
```

Results 01 / stiff_inclusions

Stiff spherical inclusions (illustrative)

Ef=70, Em=3.5 GPa; nu_f=0.2, nu_m=0.35. Data status: illustrative.



Vf	MT E (GPa)	SC E	HS lower E	HS upper E
0.05	3.8471	3.8635	3.8471	5.3749
0.1	4.2279	4.3027	4.2279	7.3088
0.3	6.2137	7.4440	6.2137	15.9603
0.5	9.4896	16.1079	9.4896	26.6571
0.6	12.0876	24.1213	12.0876	33.0487

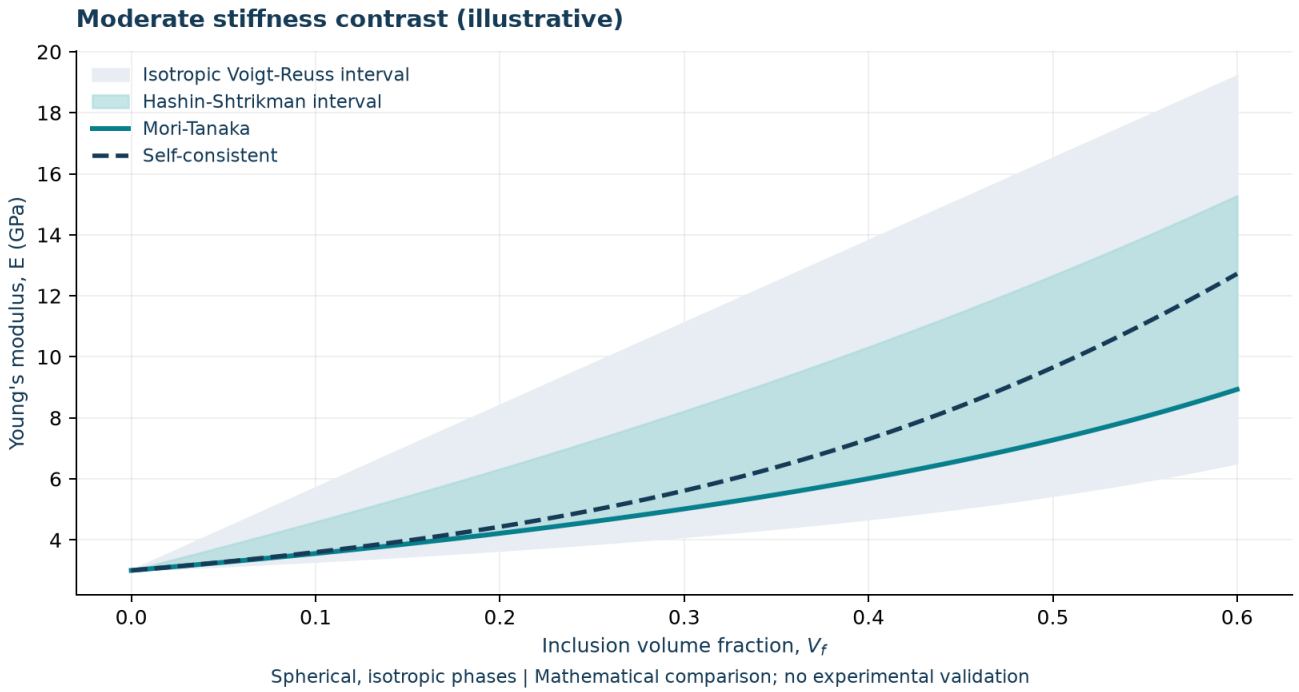
Over Vf=0 to 0.6, Mori-Tanaka changes from 3.5000 to 12.0876 GPa, a change of +245.36%. This is a model trend for the stated inputs, not a measured improvement.

Hypothetical teaching inputs, not measured glass/epoxy data. xi=2 is an uncalibrated parameter demonstration and is not a spherical-model calibration.

Results 02 / moderate_contrast

Moderate stiffness contrast (illustrative)

Ef=30, Em=3 GPa; nu_f=0.3, nu_m=0.35. Data status: illustrative.



Vf	MT E (GPa)	SC E	HS lower E	HS upper E
0.05	3.2660	3.2759	3.2660	3.7808
0.1	3.5550	3.5988	3.5550	4.5922
0.3	5.0169	5.6140	5.0169	8.2222
0.5	7.2714	9.6509	7.2714	12.6600
0.6	8.9326	12.7219	8.9326	15.2811

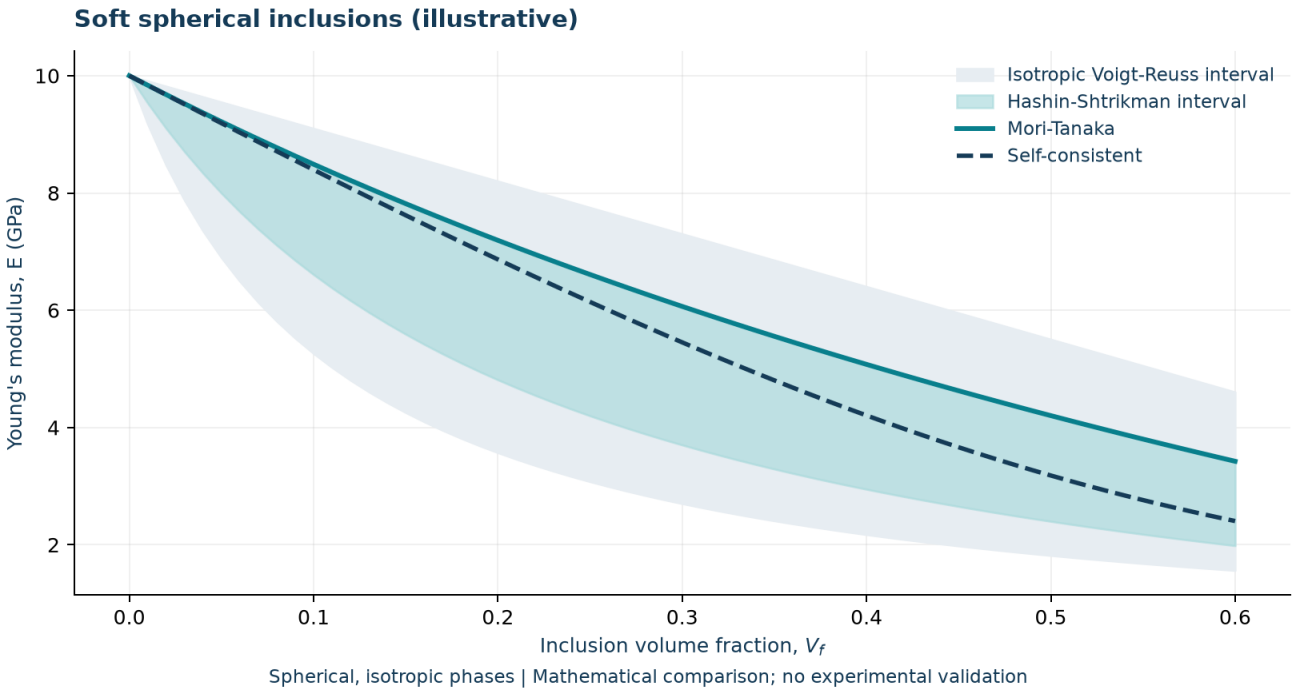
Over Vf=0 to 0.6, Mori-Tanaka changes from 3.0000 to 8.9326 GPa, a change of +197.75%. This is a model trend for the stated inputs, not a measured improvement.

Hypothetical sensitivity case, not a measured biocomposite dataset. Real natural fibers require geometry and anisotropic-property decisions. xi=2 is an uncalibrated demonstration.

Results 03 / soft_inclusions

Soft spherical inclusions (illustrative)

Ef=1, Em=10 GPa; nu_f=0.3, nu_m=0.25. Data status: illustrative.



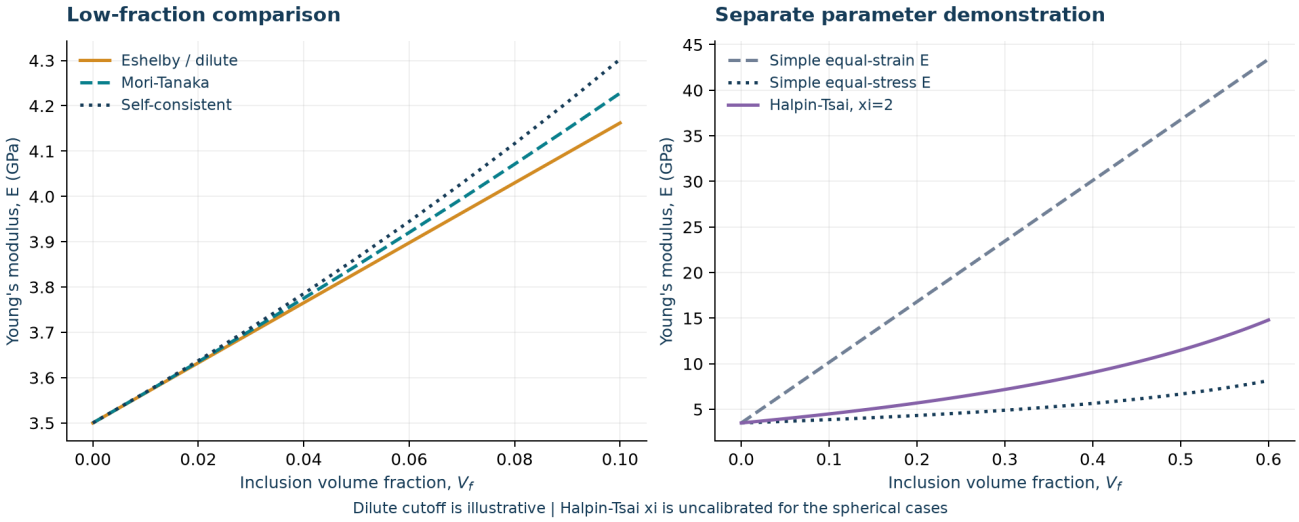
Vf	MT E (GPa)	SC E	HS lower E	HS upper E
0.05	9.2154	9.1900	7.9950	9.2154
0.1	8.4902	8.3953	6.6075	8.4902
0.3	6.0652	5.4523	3.7016	6.0652
0.5	4.2026	3.1814	2.3989	4.2026
0.6	3.4241	2.4045	1.9841	3.4241

Over Vf=0 to 0.6, Mori-Tanaka changes from 10.0000 to 3.4241 GPa, a change of -65.76%. This is a model trend for the stated inputs, not a measured improvement.

Hypothetical case to check softening and reversed phase ordering. The inclusion is a solid, not a void. xi=2 is an uncalibrated demonstration.

05 / Low-fraction and parameter demonstrations

Example shown: Stiff spherical inclusions (illustrative). Equivalent figures for every other case are included in the outputs/figures folder.



Read the panels differently

The left panel compares low-fraction isotropic estimates. Dilute and Mori-Tanaka approach one another in the dilute limit. The right panel illustrates scalar mixture and Halpin-Tsai behavior; it is not an equivalent spherical-model benchmark.

No accuracy ranking from agreement alone

A model falling between bounds is mechanically consistent with those bounds under the stated assumptions. That does not prove that it is the best prediction for a real specimen. Calibration and experimental validation are separate tasks.

06 / Verification, limitations and conclusion

The report was generated after the real test suite passed: **22 test methods**, including a fixed 120-case material sweep with independent checks. The full record is outputs/verification.txt; machine-readable status is verification.json.

Check	Purpose
Phase endpoints and identical phases	Recover known material limits
Kelvin vs Cartesian tensor mapping	Detect shear-scaling convention errors
Rational fixture and scalar reductions	Independently check tensor model calculations
SC residuals, phase exchange and units	Verify solver equations and invariances
VR/HS ordering and MT overlap	Check applicable theoretical constraints
Input, CSV and CLI integration	Check readable errors and reproducible output

Limitations

No experiments, FEM, cylindrical fibers, woven architecture, real-fiber anisotropy, voids, damage, interface slip or environmental effects are modeled. All shipped numeric material cases are illustrative. Source data, geometry decisions and validation measurements are still required before making a real biocomposite claim.

Conclusion

The project provides a reproducible analytical framework for predicting and comparing effective isotropic stiffness under explicit assumptions. It distinguishes bounds, mean-field predictions and semi-empirical demonstrations and checks their mathematical consistency.

07 / References and next study steps

R1. Echoes: Voigt/Reuss and homogenization schemes

R2. Eshelby (1957): original inclusion solution

R3. Effective moduli: spherical concentration factors (2021)

R4. NPTEL: Self Consistent Method

R5. RockPhyPy: independent HS bound formula reference

R6. NPTEL: Halpin-Tsai relations

R7. NPTEL: Composite classification

R8. NPTEL: Reinforcement materials and forms

R9. NPTEL: Weave design

Prepare for the project presentation

Explain E versus strength, why Poisson ratios are needed, the host-medium assumption of each scheme, why MT and HS may overlap, the dilute restriction and the need to justify xi. A longer question-and-answer guide is supplied in report/background_and_viva.md.

The references support theory and implementation checks. They are not sources for the hypothetical input numbers. No institutional format, student identity, experimental results or supervisor approval is invented.